

概率图模型

1. 核心理想: 解决高维度量计算困难的方法.

① 朴素贝叶斯: $P(x_1, x_2, \dots, x_n | Y) = \prod_{i=1}^n P(x_i | Y)$

② 马尔可夫性质: 未来状态仅依赖于当前, 与过去无关

③ 条件独立性: 给定某度量集合, 另两度量独立

2. 贝叶斯网络

2.1 DAG: 结点: 随机度量, 边: 直接依赖.

因子分解: $P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | pa(x_i))$

2.2. 有向分离: 判定给定观测 Z 时, X 与 Y 是否条件独立.

1. 顺连: $X \rightarrow Z \rightarrow Y$ Z 被观测, $X \perp Y | Z$

2. 分连: $X \leftarrow Z \rightarrow Y$ Z 被观测, $X \perp Y | Z$.

3. 汇连: $X \rightarrow Z \leftarrow Y$ Z 未观测, $X \perp Y$
 Z 被观测, $X \not\perp Y | Z$ (解解消除).

证: 1. $P(X, Y, Z) = P(X)P(Z|X)P(Y|Z)$.

$$P(X, Y | Z) = \frac{P(X)P(Z|X)P(Y|Z)}{P(Z)} = P(X|Z)P(Y|Z) \quad \text{条件独立} \quad \text{此时 } X \perp Y | Z.$$

2. $P(X, Y, Z) = P(Z)P(X|Z)P(Y|Z)$.

$$P(X, Y | Z) = \frac{P(X, Y, Z)}{P(Z)} = P(X|Z)P(Y|Z) \quad X \perp Y | Z$$

3. $P(X, Y, Z) = P(X)P(Y)P(Z|XY)$.

$P(X, Y) = P(X)P(Y)$, 天然独立, 因此, $X \perp Y$.

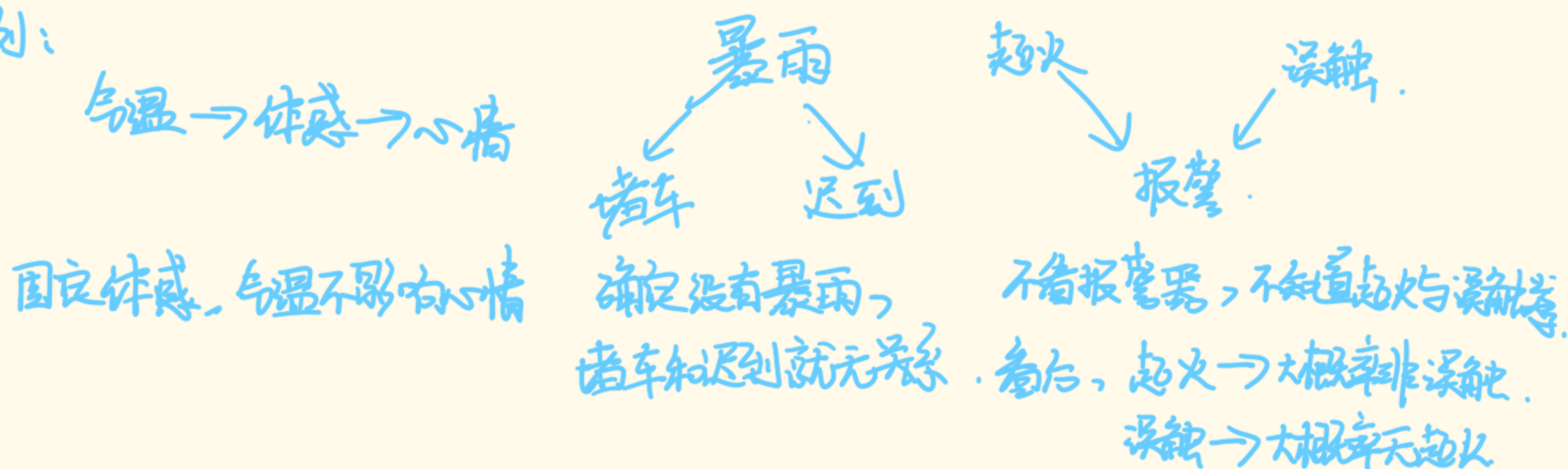
$$P(X, Y | Z) = \frac{P(X)P(Y)P(Z|XY)}{P(Z)} \quad \text{一般 } P(Z|XY) \neq P(Z), \quad X \not\perp Y | Z$$

① $X \rightarrow Z \rightarrow Y$. 观测 Z 后, X 就不可能通过 Z 影响 Y

② $X \leftarrow Z \rightarrow Y$. 观测 Z 后, X, Y 就失去了共同噪声的源头, X, Y 则独立.

③ $X \rightarrow Z \leftarrow Y$. 无 Z , X, Y 独立影响 Z ; 观测 Z , 再看 X 反而会约束 Y .

例:



在复杂图中, 两度量被阻隔 $\Leftrightarrow$ 基图上每一条路径中间度量在阻隔

2.3. 马尔可夫性:

成对: 图中任意两个无直接相连的节点, 相互条件独立.

$\wedge \forall X, Y, (X, Y) \notin E \Rightarrow X \perp Y / \text{其余节点}$

局部: 给定一个节点及节点, 其条件独立于其非后代节点.

$\wedge X \perp Nd(X) / Pa(X)$

例: $A \rightarrow B \rightarrow C \leftarrow D$ 对 C . $C \perp A / BD$.

全局: 被图 d -分离任意两组节点, 一定条件独立.

$X \text{ sep } Y / Z \Rightarrow X \perp Y / Z$ (也即上述 d -全部结论)

马尔可夫覆盖 (Markov Blanket, MB): $X \perp \text{其他节点} / MB(X)$.

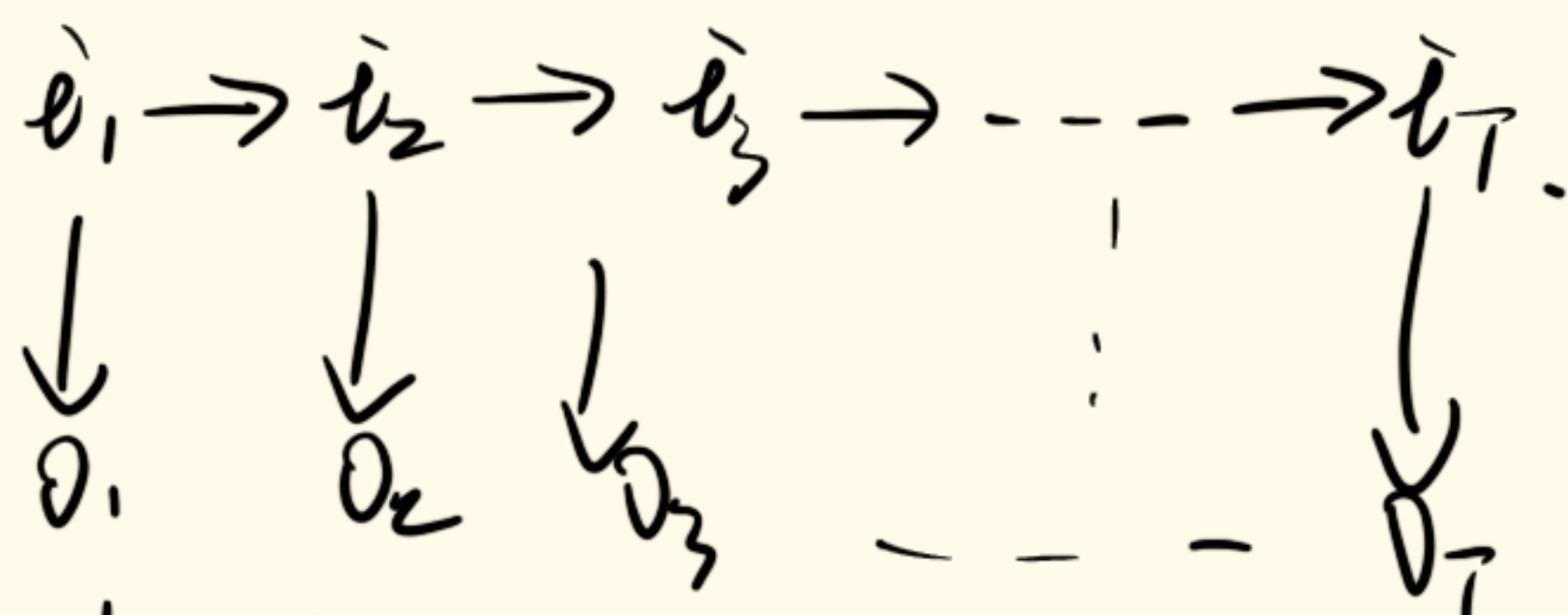
$MB(X) = \text{父} \cup \text{子} \cup \text{子节点父}$

自己父母 + 孩子 + 孩子父母.

3. 隐马尔可夫模型 (HMM)

3.1. 状态 Q , 观测 V , 初始概率 π , 状态转移矩阵 A , 观测概率矩阵 B .

隐状态序列: $i_1, i_2, \dots, i_T$. 观测序列: $o_1, o_2, \dots, o_T$



齐次一阶 Markov 假设: $i_{t+1} \perp \{i_1, \dots, i_t\} \mid i_t$

$$P(i_t \mid i_{t-1}, i_{t-2}, \dots, i_1) = P(i_t \mid i_{t-1})$$

观测独立假设: $o_t \perp \text{其他所有} \mid i_t$

$$P(o_t \mid i_t, i_{t-1}, \dots, i_1, o_{t-1}, \dots, o_1) = P(o_t \mid i_t)$$

状态集合 $Q = \{q_1, \dots, q_N\}$. DAG 上排节点 i_t 取值空间.

观测集合 $V = \{v_1, \dots, v_M\}$. DAG 下排节点 o_t 取值空间.

初始概率向量 $\pi = [\pi_1, \dots, \pi_N]$ $\pi_i = P(i_1 = q_i)$.

最开始处于各个隐藏状态的概率.

状态转移矩阵 A : $a_{ij} = P(i_{t+1} = q_j \mid i_t = q_i)$ $A = [a_{ij}]_{N \times N}$. $i_t \rightarrow i_{t+1}$ 条件概率.

观测概率矩阵: $b_j(k) = P(o_t = v_k \mid i_t = q_j)$ $B = [b_j(k)]_{N \times M}$. $i_t \rightarrow o_t$ 条件概率.

3.2. 评估问题

$$\lambda = (\pi, A, B, Q, V).$$

$O = o_1, o_2, \dots, o_T$ (一段观测序列).

求 $P(O \mid \lambda)$. 该 O 在这个 HMM 下出现的总概率.

前向概率:

$$\alpha_t(i) \triangleq P(o_1, o_2, \dots, o_t, i_t = q_i | \lambda).$$

时刻 t 处于隐藏状态 q_i , 且生成 $o_1, \dots, o_t$ 的联合概率

$$\textcircled{1} \alpha_1(i) = \pi_i b_i(o_1), \quad (t=1).$$

$$\textcircled{2} \alpha_{t+1}(j) = \left[\sum_{i=1}^N \alpha_t(i) a_{ij} \right] \cdot b_j(o_{t+1})$$

$$\textcircled{3} t=T \text{ 时, 终止状态, } P(O|\lambda) = \sum_{i=1}^N \alpha_T(i)$$

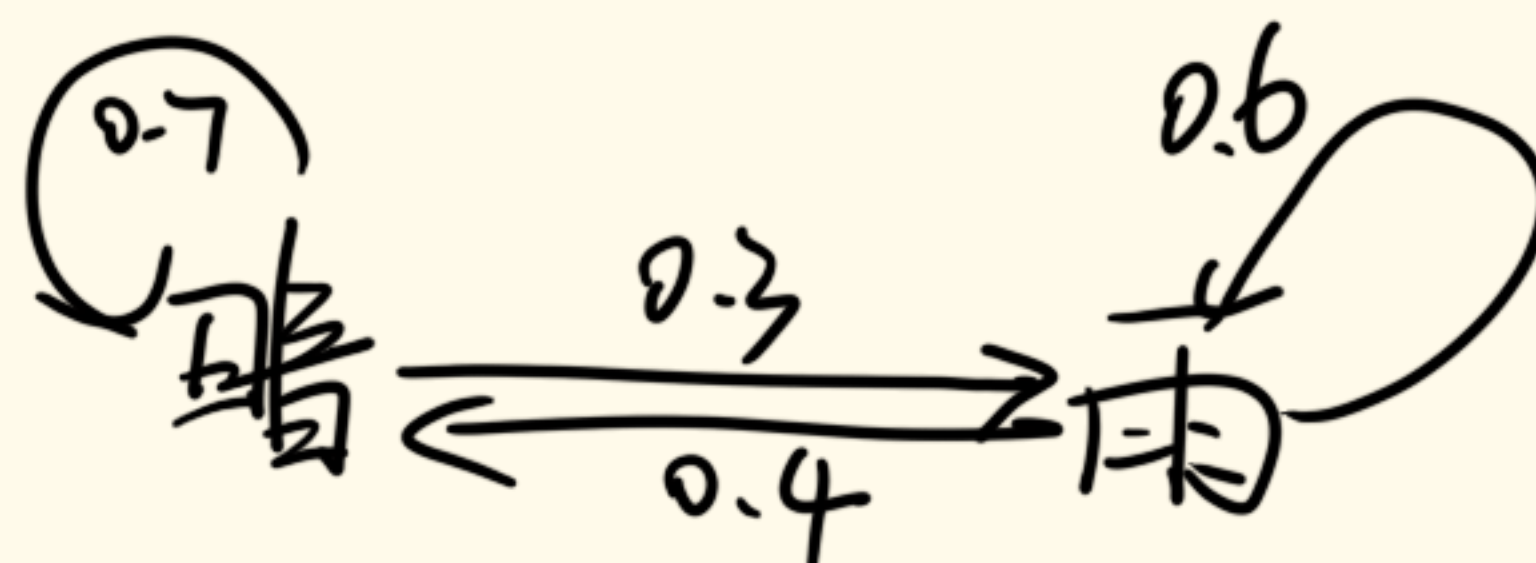
例: 设 $N=2$, $q_1 = \text{晴}$ $q_2 = \text{雨}$

$M=3$, 干、湿、潮

$$\pi = [\pi_1, \pi_2] = [0.6, 0.4].$$

$$A = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

$$B = \begin{bmatrix} \text{干} & \text{湿} & \text{潮} \\ 0.5 & 0.3 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix} \begin{matrix} \text{晴} \\ \text{雨} \end{matrix}$$



设 $T=3$, $O = (o_1 = \text{干}, o_2 = \text{潮}, o_3 = \text{湿})$

$$1. t=1. \alpha_1(i) = \pi_i b_i(o_1)$$

$$\alpha_1(1) = \pi_1 b_1(o_1) = 0.6 \times 0.5 = 0.3$$

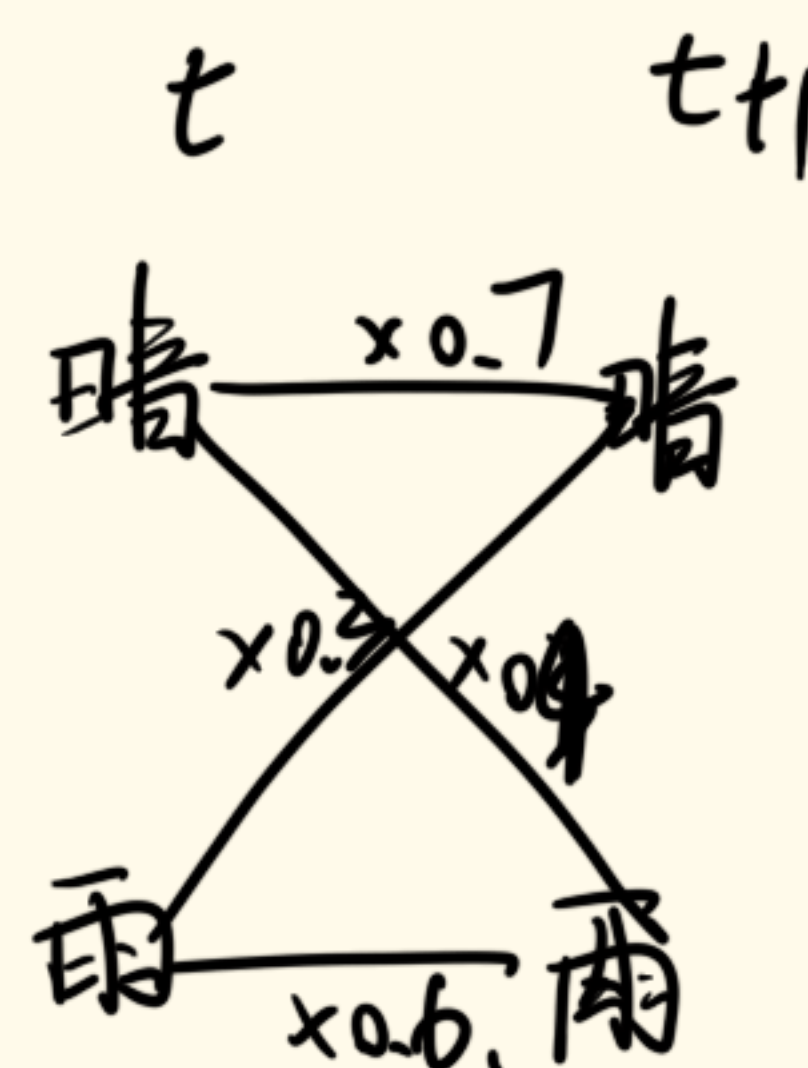
$$\alpha_1(2) = \pi_2 b_2(o_1) = 0.4 \times 0.1 = 0.04$$

$$2. t=2. \sum_{i=1}^2 \alpha_1(i) a_{i1} = 0.3 \times 0.7 + 0.04 \times 0.4 = 0.226$$

$$\sum_{i=1}^2 \alpha_1(i) a_{i2} = 0.3 \times 0.3 + 0.04 \times 0.6 = 0.114$$

$$\alpha_2(1) = 0.226 \times 0.2 = 0.0452$$

$$\alpha_2(2) = 0.114 \times 0.5 = 0.057$$



$$3. t=3. \sum_i \alpha_2(i) a_{i1} = 0.0452 \times 0.7 + 0.057 \times 0.4 = 0.05444$$

$$\sum_i \alpha_2(i) a_{i2} = 0.0452 \times 0.3 + 0.057 \times 0.6 = 0.04776$$

$$\begin{cases} \alpha_3(1) = 0.05444 \times 0.3 = 0.016332 \\ \alpha_3(2) = 0.04776 \times 0.4 = 0.019104 \end{cases}$$

$$P(O|\pi) = \alpha_3(1) + \alpha_3(2) = 0.035436.$$

我感觉这个求和写法太SB了.



我们定义 $D_t \triangleq \text{diag}(b_1(o_t), b_2(o_t), \dots, b_N(o_t))$

$$\alpha_1 = \pi^T D$$

$$\alpha_{t+1} = \alpha_t A D_{t+1} \quad P(O|\pi) = \|\alpha_T\|_1$$

本题中. $D_1 = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.1 \end{bmatrix}$ $D_2 = \begin{bmatrix} 0.2 & 0 \\ 0 & 0.5 \end{bmatrix}$ $D_3 = \begin{bmatrix} 0.3 & 0 \\ 0 & 0.4 \end{bmatrix}$

后向算法:

$$O = o_1, o_2, \dots, o_T, \text{ 求 } P(O|\pi)$$

设后向概率 $\beta_t(i)$: 已知第 t 状态为 q_i , 未来从 $t+1$ 到末尾所有观测出现的概率

当前状态 $\rightarrow$ 转移到下一时刻所有状态 $\rightarrow$ 下一时刻观测 + 未来总概率

$$P(O|\pi) = \sum_{i=1}^n \pi_i b_i(o_1) \beta_i(1) \quad \text{SB!}$$

例:

$$\pi = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} \quad A = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \quad B = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

$O = \{\text{下雨, 潮湿}\}$

$T=3$

$$\beta_T = 1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \beta_t = A D_{t+1} \beta_{t+1} \quad P = \alpha, \beta,$$

$$\beta_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\beta_2 = A D_2 \beta_3 = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \begin{bmatrix} 0.3 & 0 \\ 0 & 0.4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.33 \\ 0.36 \end{bmatrix}$$

$$\beta_1 = A D_2 \beta_2 = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \begin{bmatrix} 0.2 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 0.33 \\ 0.36 \end{bmatrix} = \begin{bmatrix} 0.1002 \\ 0.1344 \end{bmatrix}$$

$$P(O|D) = \alpha, \beta_1 = \begin{bmatrix} 0.3 & 0.04 \end{bmatrix} \begin{bmatrix} 0.1002 \\ 0.1344 \end{bmatrix} = 0.035436.$$

3.2 解码问题.

Viterbi 向量: δ_t : t 时处于某状态, 走到当前的单条最优最大概率路径.

$$\delta_1 = \pi D_1$$

δ_t : 走到第 t 步, 状态 的最大概率路径概率.

$$\delta_{t+1} = \max_{row} (\delta_t A) \odot D_{t+1}.$$

ψ : 记录上一个状态由何状态而来.

仍以本题为例: $t=1 \quad \delta_1 = \pi D_1 = \begin{bmatrix} 0.3 & 0.04 \end{bmatrix} \quad \psi_1 = [\psi, \emptyset]$

$$\delta_1 A = \begin{bmatrix} 0.3 & 0.04 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} = \begin{bmatrix} 0.226 & 0.114 \end{bmatrix}.$$

$$\psi_1: \max(0.3 \times 0.7, 0.04 \times 0.4) = 0.21 \quad |$$

$$\psi_2: \max(0.3 \times 0.3, 0.04 \times 0.6) = 0.09 \quad |$$

$$\text{故 } \psi_2 = [1 \ 1]$$

$$\delta_2 = \delta_1 A D_2 = [0.0452 \ 0.057]$$

$$\delta_2 A = [0.0452 \ 0.057] \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

$$\psi_3 = [1 \ 2]$$

$$\operatorname{argmax}(\delta_3) = 2 \quad i_3^* = 2. \quad i_2^* = \psi_3(2) = 2 \quad i_1^* = \psi_2(2) = 1.$$

$$\text{故 } I^* = (1, 2, 2) \quad \text{晴} \rightarrow \text{雨} \rightarrow \text{雨}$$

3.3. 学习问题 (Baum-Welch).

要求解 $\pi^* = (\pi, A, B)$, 已知观测, 状态数, 观测种类数.

初始化: 给定 π_0, A_0, B_0 (随机/经验).

上步: 用当前模型, 前向算 $\alpha_1, \dots, \alpha_T$

后向算 $\beta_1, \dots, \beta_T$.

$$\text{每个时刻下, } \gamma_t = \frac{\alpha_t \theta \beta_t^T}{\alpha_t \beta_t} \quad \text{状态出现的期望.}$$

$$\xi_t = \frac{\operatorname{diag}(\alpha_t^T) A D_{t+1} \beta_{t+1}}{\alpha_t \beta_t} \quad \text{转移发生的期望.}$$

$$M_{\pi}: \pi^{\text{new}} = \gamma_1$$

$$A_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}$$

$$B_{jk}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \gamma_t(i) \mathbb{1}_{\{o_t = v_k\}}}{\sum_{t=1}^{T-1} \gamma_t(i)}$$

回顾: EM 原理.

观测数据: X (已知).

隐变量向量: Z (不可见).

模型参数 θ

目标: $\max_{\theta} \log p(X; \theta) = \log \int p(X, Z; \theta) dZ.$

E步: 固定 θ_k , 求期望.

$$Q(\theta, \theta_k) = E_{p(Z|X; \theta_k)} [\log p(X, Z; \theta)].$$

M步: $\theta_{k+1} = \arg \max_{\theta} Q(\theta, \theta_k).$

在HMM中, $Q(\theta, \theta_k) = I^T (p(Z|X; \theta_k) \odot \log p(X, Z; \theta)).$

$$Z = I = [i_1, i_2, \dots, i_T].$$

$$p(\theta, I; \pi) = \pi \left(\prod_{t=1}^T D_t \right) \prod_{t=1}^{T-1} A_t.$$

E步: $p(Z|\theta; \pi_k) \Leftrightarrow \gamma_t, \xi_t$

M步: $\pi_{k+1} = \arg \max_{\pi} I^T (p(I|\theta; \pi_k) \odot \log p(\theta, I; \pi))$

解析解导出 $\pi^{new}, A^{new}, B^{new}$

整理:

前向:

$$\alpha_1 = \pi D_1,$$

$$\alpha_{t+1} = \alpha_t A D_{t+1}$$

$$P(\omega|\pi) = \|\alpha_T\|,$$

后向: $\beta_T = [1]$

$$\beta_t = A D_{t+1} \beta_{t+1}$$

$$P(\omega|\pi) = \alpha_1 \beta_1 \\ = \pi D_1 \beta_1$$

维特比算法: $\delta_1 = \pi D_1$

$$\delta_{t+1} = J D_{t+1} \quad J(\bar{i}) = \max_{i=1, \dots, N} (\delta_t(i) a_{ij}) \quad J \in \mathbb{R}^{1 \times N}$$

(对每一列取最大值)

Baum-Welch 算法.

$$E: \gamma_t = \frac{\text{diag}(\alpha_t) \beta_t}{\alpha_t \beta_t}$$

$$\xi_t = \frac{\text{diag}(\alpha_t) A D_{t+1} \beta_{t+1}}{\alpha_t \beta_t}$$

$$M: \pi_{\text{new}} = \gamma_1$$

$$\Xi = \sum_{t=1}^{T-1} \xi_t, \quad G = \text{diag}\left(\sum_{t=1}^T \gamma_t\right) A_{\text{new}} = G^{-1} \Xi$$

$$B_{\text{new}} = G^{-1}$$

$$\pi, A, B \leftarrow \pi_{\text{new}}, A_{\text{new}}, B_{\text{new}}$$

4. 马尔可夫随机场.

4.1 Hammersley-Clifford 定理:

$G=(V, E)$ 无向, P 是定义在 X 上的 严格 正分布.

P 满足 G 上的全局马尔可夫性 $\iff P$ 可以表示为 G 上的吉布斯分布

$$P(x) = \frac{1}{Z} \prod_{c \in C(G)} \psi_c(x_c)$$

4.2 吉布斯分布:

$$P(x) = \frac{1}{Z} \prod_{C \in \mathcal{C}} \psi_C(x_C)$$

$\mathcal{C}$: 图中极大团 $\psi_C(x_C)$ 势函数.

Z : 归一化常数 $Z = \sum_x \prod_{C \in \mathcal{C}} \psi_C(x_C)$.

例: $X_1 - X_2 - X_3$.

$$\mathcal{C}_1 = \{1, 2\}, \quad \mathcal{C}_2 = \{2, 3\}.$$

由 H-C 定理: $P(x_1, x_2, x_3) = \frac{1}{Z} \psi_{12}(x_1, x_2) \psi_{23}(x_2, x_3)$.

$$= \frac{\psi_{12}(x_1, x_2) \psi_{23}(x_2, x_3)}{\sum_{x_1, x_2, x_3} \psi_{12}(x_1, x_2) \psi_{23}(x_2, x_3)},$$

标准分布中, $\psi_C(x_C) = e^{-E_C(x_C)}$ $\uparrow$ x_1, x_2, x_3 的合法取值

玻尔兹曼分布: $P(x) = \frac{1}{Z} \exp\left(-\frac{E(x)}{T}\right)$ 固定能量下, 熵最大的分布.

$$Z = \sum_x \exp\left(-\frac{E(x)}{T}\right)$$

$$\max_P H_P = -\sum P(x) \ln P(x)$$

$$\text{s.t. } \sum P = 1, \quad \sum P(x) E(x) = \bar{E}$$

$$\text{解: } L = -\sum_x P(x) \ln P(x) + \alpha (\sum_x P(x) - 1) + \beta (\sum_x P(x) E(x) - \bar{E})$$

$$\frac{\delta L}{\delta P(x)} = -\ln P(x) - 1 + \alpha + \beta E(x) = 0.$$

$$p(x) = e^{\alpha-1} e^{-\beta E(x)} \\ = \frac{1}{Z} e^{-\beta E(x)}$$

最大熵模型: $L = -\sum_x p(x) \ln p(x) + \lambda_0 (1 - \sum_x p(x))$
 $+ \sum_i \lambda_i (\sum_x p(x) f_i(x) - \bar{f}_i)$

$$\frac{\partial L}{\partial p(x)} = -1 - \ln p(x) - \lambda_0 + \sum_i \lambda_i f_i(x) = 0$$

$$p(x) = \exp(-1 - \lambda_0 + \sum_i \lambda_i f_i(x)) \\ = \frac{1}{Z} e^{\sum_i \lambda_i f_i(x)}$$

此处 $E_p[f_i(x)]$
 $= \sum_x p(x) f_i(x)$

条件随机场CRF

输入序列: $x = (x_1, x_2, \dots, x_T)$

$y = (y_1, y_2, \dots, y_T)$

无向图: $y_1 - y_2 - \dots - y_T$

$$p(y|x) = \frac{1}{Z(x)} \exp(-E(y|x))$$

$$E(y|x) = \sum_{i=1}^I \psi(y_i, x) + \sum_{i=1}^{I-1} \psi(y_i, y_{i+1}, x)$$

最大熵模型 $P(y|x) = \frac{1}{Z(x)} \exp(\sum_i \lambda_i f_i(x, y))$ 判别式模型.

对数线性模型: $\log P(y|x) = \sum_i \lambda_i f_i(x, y) - \log Z(x)$

直接数学期望.

逻辑回归: 二分类下的

$$P(y=1|x) = \frac{1}{1 + \exp(-w^T x)} \quad \text{当 } f_i(x) = x \text{ 时成立}$$

$$\text{设 } y \in \{0, 1\}, \text{ 令 } f(x, y) = \begin{cases} w^T x, & y=1 \\ 0, & y=0 \end{cases}$$

$$P(1|x) = \frac{1}{Z(x)} \exp(w^T x) P(0|x) = \frac{1}{Z(x)} \exp(0) = \frac{1}{Z(x)}.$$

$$Z(x) = \exp(w^T x) + 1$$

$$P(1|x) = \frac{\exp(w^T x)}{1 + \exp(w^T x)}.$$

$$\text{对数} \quad \log \frac{P(1|x)}{P(0|x)} = w^T x.$$